

The Safety Filter: A Unified View of Safety-Critical Control in Autonomous Systems

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Keywords

safe autonomy, robot learning, safe learning, robust control, supervisory control, reinforcement learning, learning-based control, runtime assurance

Abstract

Recent years have seen significant progress in the realm of robot autonomy, accompanied by the expanding reach of robotic technologies. However, the emergence of new deployment domains brings unprecedented challenges in ensuring safe operation of these systems, which remains as crucial as ever. While traditional model-based safe control methods struggle with generalizability and scalability, emerging data-driven approaches tend to lack well-understood guarantees, which can result in unpredictable catastrophic failures. Successful deployment of the next generation of autonomous robots will require integrating the strengths of both paradigms. This article provides a review of safety filter approaches, highlighting important connections between existing techniques and proposing a unified technical framework to understand, compare, and combine them. The new unified view exposes a shared modular structure across a range of seemingly disparate safety filter classes and naturally suggests directions for future progress toward more scalable synthesis, robust monitoring, and efficient intervention.

1. INTRODUCTION

The rapid advancement of robotics and autonomous systems calls for rigorous safe control methods to ensure their reliable operation across a wide range of environments. Recent years have witnessed significant progress in the development of computational tools and theoretical frameworks to address this imperative, ranging from model-based approaches such as control barrier functions (CBFs) (1), model predictive control (MPC) (2), and reachability analysis (3) to data-driven schemes featuring learned certificates (4), adversarial reinforcement learning (RL) (5), and self-supervised safety analysis (6). With these techniques having reached a certain degree of maturity, and new opportunities opened by ongoing advances in learning and optimization, the time is ripe to take stock of the common principles and relative strengths of existing approaches in a coherent review. Unfortunately, differences in terminology and background between subcommunities have resulted in a certain degree of fragmentation in the literature, with occasional reinvention of existing ideas under neighboring frameworks. This review is driven by the conviction that connecting these approaches under a common theory will facilitate the transfer of insights between them, reducing the duplication of work and ultimately speeding up the much-needed progress in the field as a whole. The present article aims to serve as a unifying review in three distinct ways:

1. It provides an introduction to the material for students and researchers who are new to the area of safe decision-making and control and who may be seeking a roadmap to navigate the rapidly growing volume of literature addressing this problem.
2. It distills recent advances for researchers and practitioners in the field, who may be familiar with some but not all of the approaches to safety filtering and find themselves disconcerted (and perhaps exasperated) by the absence of clearly spelled out links between various coexisting methodologies and newly proposed ideas.
3. It reflects on the state of the art for the technical community as a whole, to prompt further discussion about the key challenges that the field will need to confront going forward, as well as the most promising avenues of research to address them.

We emphasize that this article is by no means the first survey of safe control approaches. It is, however, unique in providing a unified conceptual and theoretical framework under which to identify common features and distill key ideas from the plethora of recent research advances. Surveys on specific safe control methodologies include Hamilton–Jacobi (HJ) reachability (3, 7), CBFs (8), learning-based MPC (9, 10), stochastic MPC (11, 12), reachability analysis of neural networks (13), and safe control with learned certificates (14). Brunke et al. (15) provided a comprehensive review of safe robot control in the context of RL.

The present review is structured as follows: Section 2 discusses the general problem of safe decision-making under uncertainty, introducing key concepts, unifying terminology and notation from multiple areas, and motivating the idea of safety filters and safety–performance separation. Section 3 reviews the most important classes of safety filters (including some techniques not typically viewed as safety filters), attending to their historical evolution and organizing them into a straightforward taxonomy. Section 4 elucidates the common theoretical foundation that binds all safety filter approaches and further shows that any provably correct safety filter can be analyzed based on what we refer to as the universal safety filter theorem. The review concludes with a discussion of the state of the art and future challenges in Section 5.

2. SAFE DYNAMIC DECISION-MAKING

To discuss and compare the many safety filter approaches, we first establish in this section a common technical language to describe them.

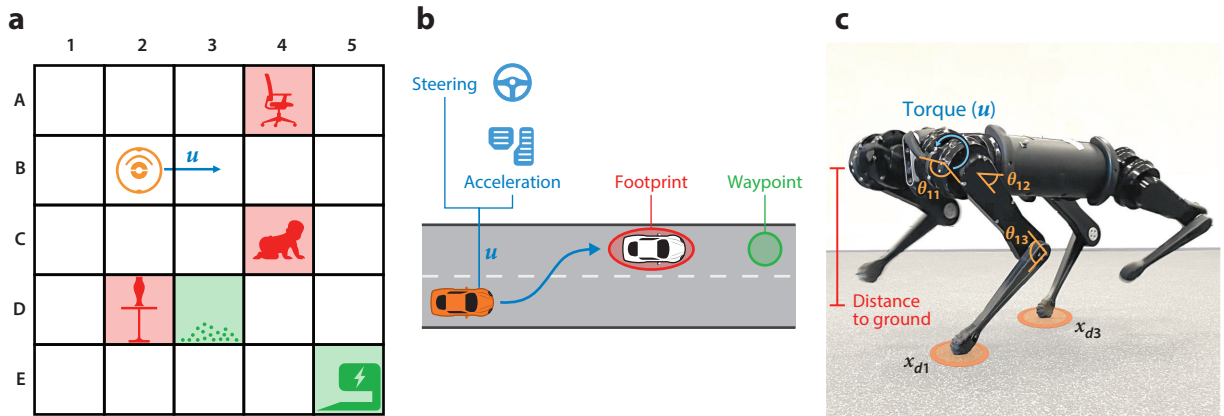


Figure 1

Examples of safety-critical autonomous systems. (a) A robot vacuum cleaner navigating an environment represented as a two-dimensional grid. (b) An autonomous car driving on a highway. (c) A quadrupedal robot walking in an indoor environment.

2.1. System Model

We consider an autonomous robot operating in a possibly dynamic environment (**Figure 1**). We denote by $x \in \mathcal{X} \subseteq \mathbb{R}^{n_x}$ the state of the overall system (i.e., the robot and its environment), a collection of variables that suffices to characterize and predict its behavior over time.

2.1.1. Dynamic modeling and predictive uncertainty. The evolution of the state x may be influenced by certain variables that the robot can adjust instantly at each control cycle, termed control inputs (or actions) $u \in \mathcal{U} \subseteq \mathbb{R}^{n_u}$, where the set $\mathcal{U}$ usually captures actuator limits (e.g., motor torque). The system is, in general, also subject to a disturbance input, denoted by $d \in \mathcal{D}$, which captures the model's predictive uncertainty, including exogenous factors (e.g., wind), changes in operating conditions (e.g., motor wear), and overall modeling error.¹ Continuous-time dynamics, $t \in \mathbb{R}$, naturally describe physical phenomena from first principles (e.g., Newton's law), whereas discrete-time models, $k \in \mathbb{Z}$, match the finite control rate at which robots compute and execute decisions. Since most safe control frameworks lend themselves to both paradigms, we adopt the discrete-time formulation for brevity, while cautioning that, when reasoning with discrete-time models, it is important in practice to ensure that safety is satisfied in between the time steps.² The system's evolution in discrete time can be described by

$$x_{k+1} = f(x_k, u_k, d_k). \quad 1.$$

Note that an uncertain dynamical function $f : \mathcal{X} \times \mathcal{U} \times \mathcal{D} \rightarrow \mathcal{X}$ (and an associated set $\mathcal{D}$) always exists for any recorded system history $(x_0, u_0, x_1, u_1, \dots)$, such that Equation 1 is satisfied at each time k for some value $d_k \in \mathcal{D}$.³

2.1.2. Sensor modeling and perceptual uncertainty. While the assumption of perfect state observability—always a simplification—is practically acceptable in many settings, there are

¹It is not that the function describing the evolution of the modeled x given u over time cannot be computed exactly, but rather that such a “true” function may not exist at all. There may be two instances in which the same initial state and control signal result in different state trajectories.

²Although not always addressed explicitly, the translation of discrete-time safety certificates into continuous-time guarantees has been studied using sampled-data system theory (e.g., 16).

³A trivial, extreme example is the agnostic model $f(x_k, u_k, d_k) := d_k$, with $d_k \in \mathcal{D} \equiv \mathcal{X}$.

Control rate: the frequency (typically expressed in hertz) with which a robot can update its control input by executing a complete control cycle

important cases in which it is not. Significant perceptual uncertainty can result from, for example, sensor noise and occlusion. The robot receives uncertain measurements modeled as

$$y_k = b(x_k, u_k, d_k), \quad 2.$$

analogous to the uncertain dynamics in Equation 1. For conciseness, we let a single disturbance variable $d_k \in \mathcal{D}$ encode uncertainty in both dynamics and perception. When the state x_k cannot be directly observed, the robot's decisions instead rely on partial knowledge, represented in terms of an information state $\eta_k \in \mathcal{H}$ that encodes the robot's initial state knowledge η_0 and all subsequent observations $(y_1, \dots, y_k)$. In many cases, η_k can be parameterized more succinctly without negatively impacting the robot's decision-making ability. Under probabilistic uncertainty, for example, this sufficient statistic takes the form of a belief state $b_k \in \Delta(\mathcal{X})$, a distribution over the current state. Robust formulations, on the other hand, encode η_k as a set $\tilde{\mathcal{X}}_k \subseteq \mathcal{X}$ of possible states. The dynamics of η_k depend on Equations 1 and 2 but can be succinctly expressed as $\eta_{k+1} = f^\eta(\eta_k, u_k, d_k)$.

In practice, to reduce the overall state dimensionality, a number of methods work with mixed observability, i.e., a joint state (x_k, η_k) combining perfectly observable state variables and an information state over uncertain ones (e.g., 12, 17, 18). Consistently with the bulk of the safe control literature, we adopt the perfect state assumption through much of the article's exposition, emphasizing the differences with imperfect information where relevant. Safety filters for limited perception settings are discussed in detail in Section 3.6.

2.2. Safety Specifications

Given a (possibly uncertain) system model, safety specifications allow designers to encode critical requirements on system behavior and the terms under which these must be enforced.

2.2.1. Failure conditions. In many robotic applications, there exist potential failure conditions that are deemed unacceptable and must be avoided at all costs. For example, autonomous vehicles should never drive off the highway, robotic prostheses should never force their users' joints past their range of motion, and drones should never fall out of the sky. These systems are called safety-critical: Maintaining safety is a hard requirement, not merely a desirable feature to be traded off with other performance metrics. These catastrophic failure conditions can be concisely encoded by a failure set $\mathcal{F} \subseteq \mathcal{X}$ that the state is forbidden from entering. That is, the robot must always enforce the state constraint $x \notin \mathcal{F}$, or equivalently $x \in \mathcal{F}^c$, where the complement $\mathcal{F}^c := \mathcal{X} \setminus \mathcal{F}$ is called the constraint set.

2.2.2. All-time safety and controlled-invariant sets. While any given autonomous mission takes a finite amount of time, its duration usually exceeds the robot's explicit planning and prediction horizon. As a result, we typically require the robot to ensure safety for all future times—that is,

$$x_k \notin \mathcal{F}, \quad \forall k \geq 0. \quad 3.$$

Naturally, the robot cannot explicitly check an infinite sequence of time steps. Instead, all-time safety can be implicitly guaranteed through the property of set invariance. For now, we focus on the fully observable case ($y \equiv x$). A set $\Omega \subseteq \mathcal{X}$ is controlled-invariant under the dynamics in Equation 1 if there exists a control policy $\pi : \mathcal{X} \rightarrow \mathcal{U}$ such that

$$\forall x \in \Omega, f(x, \pi(x), 0) \in \Omega. \quad 4.$$

In other words, the robot can prevent the system from ever leaving the set by continually applying policy π . This deterministic definition ($d = 0$) can readily be extended to robust or

Information state:

a dynamic variable that encodes all relevant knowledge that is available for decision-making at a given time

Sufficient statistic:

a compact representation of data that contains all the necessary information for decision-making

Belief state:

a conditional distribution over the system state $b_k := p(x_k | \mathbf{y}_k)$, where $\mathbf{y}_k$ is the observation history up to time k

Robust controlled-invariant set:

a set $\Omega \subset \mathcal{X}$ is robustly controlled-invariant under bounded uncertainty $d \in \mathcal{D}$ if there is a control policy $\pi : \mathcal{X} \rightarrow \mathcal{U}$ that enforces $f(x, \pi(x), d) \in \Omega$ for any $x \in \Omega$ and for all values $d \in \mathcal{D}$

SAFE SET VERSUS CONSTRAINT SET

Note that the notion of a safe set Ω is not equivalent to the state constraint set $\mathcal{F}^c$, and in fact the set $\mathcal{F}^c$ is not in general a safe set. There usually exist states that are not explicit failure conditions but from which the robot cannot prevent a future safety violation, and they are therefore unsafe states. For example, a drone not currently in collision is still unsafe if it is flying toward an obstacle too fast to avoid it.

SAFETY VERSUS STABILITY

While related, safety and stability are distinct properties, and neither is in general sufficient for the other (a system may be safe but not stable, or stable but not safe). Certain stability methods, such as Lyapunov analysis, allow a forward-invariant region of attraction to be established around an equilibrium state; if this region happens to be disjoint from the failure set $\mathcal{F}$, it is by definition a safe set Ω . Stability analysis tools are therefore often used to construct sufficient conditions for all-time safety.

probabilistic controlled-invariant sets in the presence of disturbances.⁴ If the controlled-invariant set Ω is disjoint from the failure set, $\Omega \cap \mathcal{F} = \emptyset$, then it is called a safe set (see the sidebars titled Safe Set Versus Constraint Set and Safety Versus Stability).

In most cases, finding some safe set Ω is not particularly difficult; however, finding one large enough to be of practical utility is less straightforward. In a static environment, for example, the set of collision-free at-rest robot configurations is a safe set for collision avoidance purposes: The robot can remain in this set, and thereby collision free, by never moving—although of course, some hard-to-please customers may return the robot on the basis that it doesn't do anything. Less extreme (but similarly impairing) examples were common in the early years of social robot navigation (see the freezing robot problem in Reference 19) and even autonomous driving technology, where prototype vehicles would comically settle for the off-ramp after failing to secure a safe merge onto a moderately busy highway (20). The need to ensure safety without needlessly hindering the robot's operation (further discussed in Section 2.3) motivates the notions of maximal safe set Ω^* and optimal safety policy π^* . That is, from what states is it possible for the robot to maintain safety, and how?

2.2.3. Safety in the face of uncertainty: robust versus probabilistic guarantees. Robust safety formulations aim to find a control strategy for the robot that can guarantee safety for all possible realizations of the uncertainty within a bounded (although usually infinite) set. That is, the robot's policy $\pi : \mathcal{X} \rightarrow \mathcal{U}$ should enforce the robust safety condition

$$x_k \notin \mathcal{F}, \quad \forall k \geq 0, \quad \forall d_0, d_1, \dots, d_k \in \mathcal{D}. \quad 5.$$

In contrast, probabilistic safety formulations aim to satisfy a chance constraint under a given distribution of uncertainty realizations (typically $d_k \sim p_d$ independent and identically distributed). Policy π then enforces

$$P(x_k \notin \mathcal{F}, \forall k \geq 0) \geq 1 - \delta. \quad 6.$$

In other cases, the chance constraint is instead imposed at each time step (marginally) or for a finite horizon $k \in \{0, 1, \dots, H\}$. Note that the chance constraint of Equation 6 becomes equivalent to

Probabilistic controlled-invariant set:

a set $\Omega \subset \mathcal{X}$ is probabilistically controlled-invariant with confidence level $1 - \delta \in [0, 1]$ under uncertainty $d_k \sim p_d$ if there is a control policy $\pi : \mathcal{X} \rightarrow \mathcal{U}$ that enforces $P(x_k \in \Omega, \forall k \geq 0) \geq 1 - \delta$ for any $x_0 \in \Omega$

Safe set: a region in the state space from which the robot can ensure safety at all future times

Maximal safe set: the (unique) set of all states from which it is possible for some control policy to maintain safety for all time

Optimal safety policy: a control policy that enforces all-time safety from all states from which maintaining safety is in fact possible

⁴Similarly, a controlled-invariant information set $\Omega \subseteq \mathcal{H}$ may be defined when $y \neq x$.

Operational design domain: the set of conditions in which an engineering system, particularly an automated one, has been specifically designed to operate

the robust constraint of Equation 5 when $\delta = 0$, as long as the support of the distribution p_d is $\mathcal{D}$. Conversely, for $\delta > 0$, safety may be compromised even when all modeling assumptions hold, since catastrophic failures $x_k \in \mathcal{F}$ are allowed to happen with a small nonzero probability.

Probabilistic safety guarantees are appropriate in settings with high uncertainty and limited stakes: A robot vacuum cleaner should not damage valuables or bump into toddlers (**Figure 1a**), but occasional failures may be acceptable as long as they are sufficiently rare. This contrasts with high-stakes domains like autonomous driving (**Figure 1b**), where near-zero failure probabilities can cause important practical issues. Many types of safety-critical events are unlikely due to their extreme nature and, as a result, may fall through the cracks in a probabilistic assessment, even when their handling may be crucial to the system’s viability.⁵ Achieving acceptable safety levels through statistics-informed design and evaluation is a notoriously difficult and long-standing challenge in the autonomous driving field, where it is commonly referred to as the long tail (23).

2.2.4. Operational design domain: qualifying safety. A commonly voiced objection against technical approaches seeking rigorous safety guarantees is that they aim to ensure absolute safety, thereby setting an impossible standard. This is a fundamental misinterpretation of what safety guarantees are. The wide range of principled safety frameworks—and safety filters in particular—can best be viewed as establishing conditions under which a particular system will not experience a specified class of catastrophic failures. In the last decade, the autonomous driving community has led significant efforts to formalize the notion of an engineering system’s operational design domain, a clearly delineated set of operating conditions under which it can be expected—and required—to function correctly and safely (24, 25). These may, for example, include “environmental, geographical, and time-of-day restrictions, and/or the requisite presence or absence of certain traffic or roadway characteristics” (24, section 3.21).

More broadly, we can apply this idea to any autonomous robotic system: What restrictions and assumptions need to be imposed on its operating conditions (where and how it is deployed, the range of terrain and weather it may encounter, and/or the behavior of nearby agents) in order to ensure safety? The form of these assumptions is often tied to the sought guarantees: Probabilistic safety assurances follow most naturally from statistical assumptions, while robust guarantees are more readily derived from assumed bounds on uncertain variables. In this sense, the modeling of uncertainty as either $d_k \sim p_d$ or $d_k \in \mathcal{D}$, with the associated prediction and perception models, can be viewed as establishing an operational design domain for the safety guarantees offered by different approaches. Given their practical importance, the review in Section 3 pays special attention to the assumptions and guarantees associated with different classes of safety filters.

2.3. Safety Filters: Reconciling Safety and Performance

Naturally, robotic systems should not only avoid reaching catastrophic failure states (this, by itself, can be accomplished by not turning them on) but also carry out their assigned tasks reliably and efficiently. Constraining a robotic system to remain in an overly restrictive safe set can render it impractically inefficient at its intended function, or even unable to fulfill it. The desirable robot operation is therefore one that achieves the highest possible task performance while on the other hand ensuring that safety will be maintained at all times (with the required type of guarantee and under the assumptions comprising its operational design domain). Formally, the overall goal is to

⁵The fact that a safety-critical system may fail systematically in certain low-probability conditions and still be deemed “statistically safe” underpins the ongoing regulatory investigations and public controversy around the safety standards of some automated driving systems (21, 22).

find a robot control policy $\pi : \mathcal{X} \rightarrow \mathcal{U}$ to optimize a prescribed performance objective—which depends on the state and control over time—subject to the appropriate safety condition:

$$J^*(x_0) = \min_{\pi} J(x_0, \pi) \quad 7a.$$

$$\text{subject to} \quad \text{safety condition (Equation 5 or 6).} \quad 7b.$$

Importantly, the uncertainty characterization in the performance objective J is, in general, distinct from the one in the safety condition. For example, the objective in Equation 7a may encode an expected cost with $\bar{d}_k \sim \bar{p}_d$, while the constraint in Equation 7b may consider robust safety under $d_k \in \mathcal{D}$. Finding a suitable π to jointly achieve performance and safety can prove challenging and can be cumbersome to repeat for each target task. Fortunately, the problem can be broken down.

A safety filter is an automatic process that monitors the operation of an autonomous system at runtime and intervenes, when deemed necessary, by modifying its intended control action in order to prevent a potential catastrophic failure. The controlled-invariance property of the maximal safe set Ω^* implies the existence of a perfect or least-restrictive safety filter that enforces the safety requirement while only overriding control actions that would cause the system to immediately leave Ω^* .

Proposition 1 (perfect safety filter). Consider a system with dynamics as in Equation 1 under uncertainty $d \in \mathcal{D}$ and safety requirement as in Equation 5. For each state $x \in \mathcal{X}$, let $\mathcal{U}^s(x) \subseteq \mathcal{U}$ contain all control actions that robustly keep the next system state inside the maximal safe set Ω^* . Then, there exists a safety filter $\phi : \mathcal{X} \times \mathcal{U} \rightarrow \mathcal{U}$ such that (a) $\phi(x, u) \in \mathcal{U}^s(x)$ wherever $\mathcal{U}^s(x)$ is not empty, (b) $\phi(x, u) = u$ whenever $u \in \mathcal{U}^s(x)$, and (c) the filtered system satisfies the original safety requirement. Additionally, the safe decision problem in Equation 7 can be transformed into a safety-agnostic decision problem through ϕ :

$$J^*(x_0) = \min_{\pi^{\text{task}}} J(x_0, \pi^{\text{task}}) \quad 8a.$$

$$\text{subject to} \quad u_k = \phi(x_k, \pi^{\text{task}}(x_k)). \quad 8b.$$

The above result means that it is possible to design a safety filter for a robotic system with no knowledge or consideration of the particular task objectives, only requiring the filter to be as permissive as possible while preventing any safety violations. In turn, the task-oriented control policy can then be relieved of the burden of maintaining safety, since it can be viewed as acting on a modified system shielded by the safety filter. From the task policy's perspective, this filtered system is incapable of entering the failure set, regardless of the commanded control actions. Because only unsafe actions are preempted by the filter, the robot is still able to achieve the optimal viable task performance. This decoupling between safety and performance serves as a sort of separation principle, enabling the independent design of safety-agnostic task policies and task-agnostic safety filters (see the sidebar titled Safe Learning).

Unfortunately, such a least-restrictive safety filter cannot be tractably obtained for most robotic systems of practical interest. Instead, modern safety filters seek diverse trade-offs among permissiveness, robustness, generality, and computational scalability. In the next section, we review the most prominent classes of safety filter schemes, which, despite their algorithmic differences, share a core structure and important characteristics that make it possible to compare and categorize them. Then, we establish in Section 4 a unified theoretical framework that distills this structure to reveal a universal working principle.

Performance objective:

quantification of the robot's success at a given task, typically defined as a combination of metrics accounting for factors like completion time, energy usage, and human user comfort and satisfaction

SAFE LEARNING

In recent years, the safety–performance separation afforded by safety filters has proven particularly useful in enabling safe learning schemes for robots and other safety-critical autonomous systems. Indeed, this goal has driven the development of a variety of safe RL approaches whose core element is a suitable type of safety filter ϕ that allows a conventional (safety-agnostic) learning algorithm to freely explore and iteratively update π^{task} without damaging the physical system or its environment. We direct readers to Reference 15 for a recent review of safe learning.

3. SAFETY FILTERS

The runtime operation of every safety filter can be conceptualized as two complementary functions: monitoring and intervention. The safety filter continually monitors the robot’s proposed course of action to determine whether (or to what degree) it poses a safety risk. Accordingly, the filter may intervene by modulating or completely overriding the robot’s candidate control input to ensure that safety is maintained. In many safety filters, the monitoring and intervention procedures are informed by a safety-oriented control strategy that the filter considers as a potential fallback.⁶ This fallback safety strategy may be obtained by various means, including offline and runtime computation as well as manual engineering design. Depending on the underpinnings of these components, a safety filter may offer different forms of guarantees, or none whatsoever. **Figure 2** shows a general diagram.

Safety filters can therefore be categorized on the basis of their safety monitoring methodology, their intervention scheme, the synthesis approach used to obtain the fallback safety strategy, and the safety assurances provided. **Figure 3** presents a visual summary of the main safety filter classes reviewed in this article, organized by these four characteristics.

3.1. Computing the Least-Restrictive Safety Filter: Dynamic Programming

A desirable safety filter is one that grants the robot as much freedom as possible to carry out its task while allowing no safety violations. Computing the maximal safe set Ω^* [also called the viability kernel (26)] is intrinsically an optimal decision problem. Under worst-case uncertainty, safety analysis can be viewed as a pursuit–evasion game between a controller striving to avoid the failure set and an antagonistic disturbance agent aiming to steer the system into it (27–29). The HJ reachability analysis framework uses level set representations to transform the pursuit–evasion game, conceptually formulated as a game of kind, into a game of degree, by optimizing a continuous payoff (see the sidebar titled Games of Kind and Games of Degree). Formally, in discrete time, the game’s solution can be obtained via the dynamic programming Isaacs equation (27):

$$V_k(x) = \min \left\{ g(x), \max_{u \in \mathcal{U}} \min_{d \in \mathcal{D}} V_{k+1}(f(x, u, d)) \right\}, \quad k \in \{0, 1, \dots, H-1\}, \quad V_H(x) = g(x), \quad 9.$$

where $g(x)$ is a margin function encoding the failure set $\mathcal{F}$ as its zero sublevel set; a typical choice of g is the signed distance function to $\mathcal{F}$. In the continuous-time formulation, the outcome of the differential game is given by the viscosity solution (33) to a Hamilton–Jacobi–Isaacs variational inequality that is the continuous-time counterpart of Equation 9. The safety value function $V_k(\cdot)$ in Equation 9 can be computed (up to numerical error) by backward induction in time, following the dynamic programming principle, over a compact domain of the state space, in practice often

Margin function:

a Lipschitz-continuous function $m : \mathbb{R}^n \rightarrow \mathbb{R}$ encoding a given set $\mathcal{M} \subset \mathbb{R}^n$ as its zero sublevel set, $\mathcal{M} = \{x \mid m(x) < 0\}$

Signed distance function:

for any nonempty set $\mathcal{M} \subset \mathbb{R}^n$, the signed distance function $s_{\mathcal{M}} : \mathbb{R}^n \rightarrow \mathbb{R}$ is defined as the closest distance to $\mathcal{M}$ for points $x \notin \mathcal{M}$ and the negative closest distance to $\mathcal{M}^c$ for points $x \in \mathcal{M}$ under some metric in $\mathbb{R}^n$

Viscosity solution:

a general solution concept for partial differential equations that allows the solution to not be everywhere differentiable as long as it satisfies the partial differential equation almost everywhere

⁶In fact, as discussed in Section 4, every safety filter can be viewed as relying on a fallback policy.

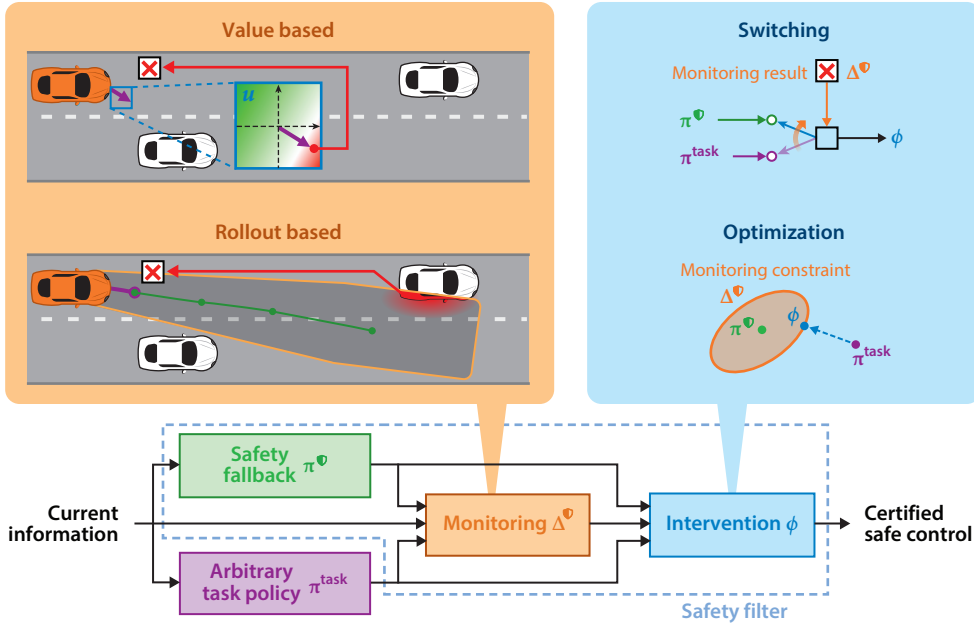


Figure 2

General diagram of a safety filter. The filter consists of three components: a safety monitor Δ^V , which determines whether the control proposed by a task policy π^{task} is safe; an intervention scheme, which modifies the candidate control as needed to satisfy the safety requirement; and a safety-oriented fallback policy π^V , which informs monitoring and intervention. Many monitoring approaches can be categorized as value based (evaluating a function that encodes whether a state or action is safe) or rollout based (computing an explicit prediction of the system's potential evolution). Common intervention approaches include switching between the task policy and the fallback strategy and optimizing the control to accommodate task performance criteria within the confines of safety. Figure adapted from Reference 5.

represented by a numerical grid. The induction converges (typically in finitely many time steps) to the infinite-horizon safety value function $V(x) = \lim_{k \rightarrow \infty} V_k(x)$. This value function encodes the maximal safe set as well as an optimal safety policy:

$$\Omega^* = \{x \in \mathcal{X} \mid V(x) \geq 0\}, \quad \pi^*(x) = \underset{u \in \mathcal{U}}{\operatorname{argmax}} \min_{d \in \mathcal{D}} V(f(x, u, d)). \quad 10.$$

The value function and policy in Equation 10 can be used to construct a safety filter with value-based monitoring and switch-type intervention (34):

$$\phi(x, \pi^{\text{task}}(x)) := \begin{cases} \pi^{\text{task}}(x), & \min_{d \in \mathcal{D}} V(f(x, \pi^{\text{task}}(x), d)) \geq 0, \\ \pi^*(x), & \text{otherwise.} \end{cases} \quad 11.$$

Intervention through $\pi^*(\cdot)$ is only needed near the boundary of Ω^* , to prevent $V(x)$ from becoming negative. While the filter is minimally restrictive in the sense of Proposition 1, control chatter and erratic motion can result near the safe set boundary due to intermittent overrides of any task policy that persistently attempts to leave Ω^* . This motivates designing intervention strategies that yield a smoother transition from performance-oriented to safety-preserving control, which we discuss further in Section 3.5.3.

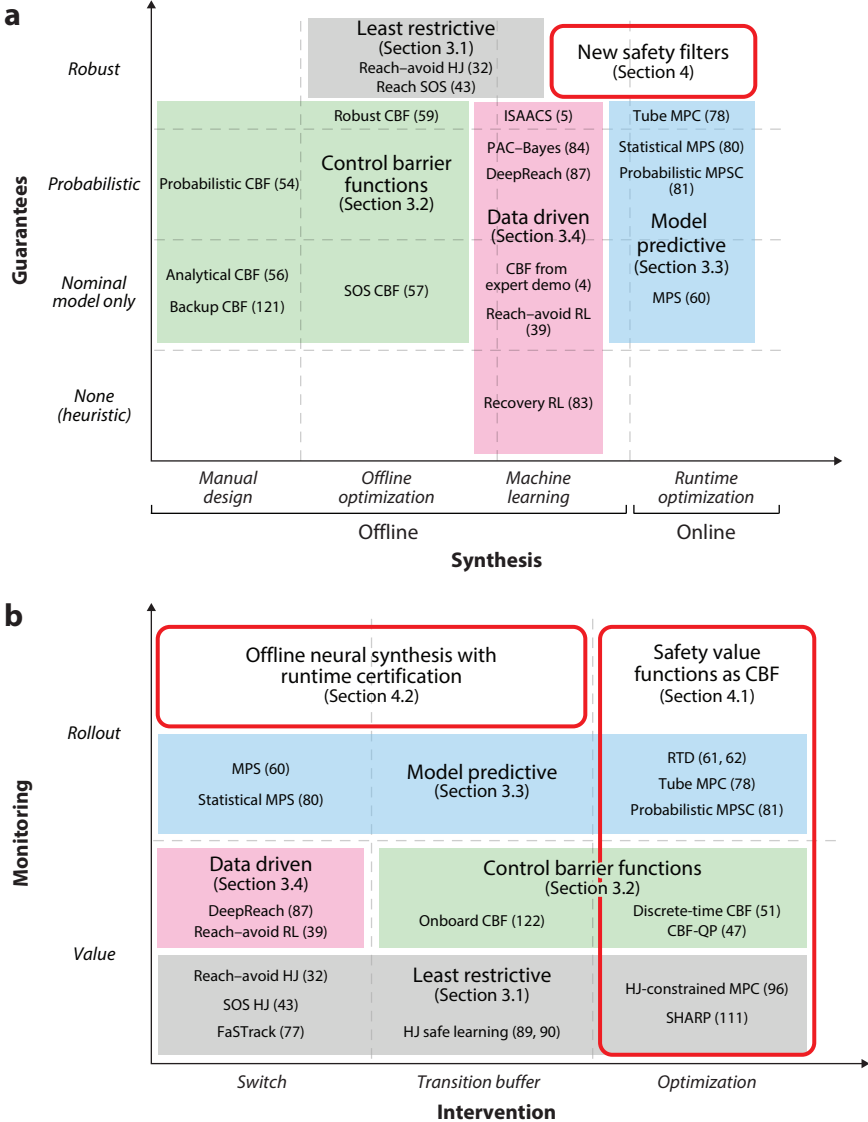


Figure 3

Summary of the safety filters reviewed in Sections 3 and 4. The representative safety filter schemes are displayed under each category. (a) Safety guarantees provided by safety filters vary according to the properties and synthesis of the underlying fallback safety strategies. (b) The runtime operation of safety filters is determined by the way they monitor the safety risk and intervene when they deem it to be unsafe. Abbreviations: CBF, control barrier function; CBF-QP, control barrier function–quadratic program; FaSTrack, fast and safe tracking; HJ, Hamilton–Jacobi; ISAACS, iterative soft adversarial actor–critic for safety; MPC, model predictive control; MPS, model predictive shielding; MPSC, model predictive safety certification; PAC, probably approximately correct; RL, reinforcement learning; RTD, reachability-based trajectory design; SHARP, shielding-aware robust planning; SOS, sum of squares; tube MPC, tube-based model predictive control.

GAMES OF KIND AND GAMES OF DEGREE

Rufus Isaacs, who worked alongside Richard Bellman at the RAND Corporation in the 1950s and 1960s, pioneered the study of differential games and their connection with reachability and safety (27). He drew a distinction between games of kind, with a categorical outcome identifying a clear winner, and games of degree, in which players instead compete over a continuum of possible outcomes. Isaacs further showed that any game of kind can be represented implicitly as a game of degree. For example, the game of whether a pursuer captures an evader can be transformed into a game in which the pursuer tries to minimize their closest future distance while the evader tries to maximize it. If the separation falls below a specified capture radius, the pursuer wins the game of kind; otherwise, the evader wins. Importantly, the optimal strategies of the game of degree are also optimal for the corresponding game of kind: Minimizing the closest distance over time is an optimal approach to ensure capture, while maximizing it is an optimal way to escape. This idea subsequently sparked numerous studies of pursuit–evasion games using level set approaches (30–32).

The computational complexity and memory requirement of numerically solving Equation 9 for general nonlinear dynamics is exponential in the state dimension, which makes the approach impractical for systems with more than six continuous state variables. To improve the tractability of HJ methods, several approaches have been explored, including system decomposition (35), efficient initializations (36), and customized hardware design (37). Moreover, approximate solutions have been achieved through techniques such as deep RL (38, 39), deep supervised learning (6, 40), iterative linear–quadratic optimization (41, 42), and sum-of-squares programming (43). Rosolia & Borrelli (44) proposed a runtime method to compute a gradually larger safe set by aggregating local model predictive fallback policies. Fridovich-Keil et al. (45) developed a similar scheme to obtain an increasingly permissive safety filter for exploration of a priori unknown environments, bootstrapping the underlying reachability analysis and fallback policy with graph search methods.

Rather than aim to tractably approximate Ω^* , multiple approaches forgo the maximal safe set and instead seek a smaller safe set $\Omega \subset \Omega^*$ that can still produce a reliable, if more restrictive, safety filter. Sections 3.2–3.4 are dedicated to reviewing these methods.

3.2. Value-Based Safety Filters: Control Barrier Functions

Initially developed by Wieland & Allgöwer (46) and later refined by Ames et al. (1, 47), CBFs retain some important properties of the safety value function $V(x)$. However, CBFs no longer encode or approximate the maximal safe set Ω^* . The concept is motivated by earlier work on control Lyapunov functions (48–50), which—if found—provide a sufficient condition to stabilize a system. An important feature of CBFs is that they allow a smooth intervention mechanism that relies on runtime control optimization.

Similar to least-restrictive filters, a CBF safety filter renders a certain state function nondecreasing on its zero level set. Formally, a (zeroing) CBF is a twice continuously differentiable function $b : \mathbb{R}^{n_x} \rightarrow \mathbb{R}$ that encodes a safe set $\Omega := \{x : b(x) \geq 0\}$ and, additionally, satisfies the following inequality regarding its rate of decrease:

$$\sup_{u \in \mathcal{U}} \nabla b(x)^\top f^c(x, u) \geq -\alpha(b(x)), \quad \forall x \in \Omega, \quad 12.$$

where $\alpha : \mathbb{R} \rightarrow \mathbb{R}$ is an extended class $\mathcal{K}$ function, and $\dot{x} = f^c(x, u)$ represents the system's continuous-time dynamics. If f^c is control-affine, i.e., $f^c(x, u) := \tilde{f}(x) + \tilde{g}(x)u$, then the inequality in Equation 12 becomes affine in control u , and a convex quadratic program [referred to as CBF-QP (1, 47)] can be solved at each state x to produce a safe control $\phi(x, u)$ that minimally

Class $\mathcal{K}$ function:

a continuous function $\alpha : \mathbb{R} \rightarrow \mathbb{R}$ is an extended class $\mathcal{K}$ function if it is strictly increasing with $\alpha(0) = 0$; a simple choice of $\alpha(b(x))$ can be $\alpha(b(x)) := b(x)$

deviates from a (potentially unsafe) task control $\pi^{\text{task}}(x)$ while satisfying the CBF decrease rate condition:

$$\phi(x, \pi^{\text{task}}(x)) = \underset{u \in \mathcal{U}}{\operatorname{argmin}} \quad \frac{1}{2} \|u - \pi^{\text{task}}(x)\|^2 \quad 13a.$$

$$\text{subject to} \quad \nabla b(x)\bar{f}(x) + \nabla b(x)\bar{g}(x)u + \alpha(b(x)) \geq 0. \quad 13b.$$

Since their debut, CBF-based safety filters have been extended to handle discrete-time problems (51), perceptual uncertainty (52, 53), and, most recently, systems with probabilistic and worst-case uncertainty (54, 55). In contrast to the efficient, systematic procedure for computing safe control actions given a known CBF, finding a valid CBF for a system of interest is often a nontrivial problem, and the search for computationally tractable and data-efficient CBF synthesis methods remains an open research challenge. Recent efforts have explored analytical design (56), sum-of-squares synthesis (57), and learning from expert demonstrations (4, 58, 59). Unfortunately, existing CBF synthesis approaches tend to work only for restricted dynamical system classes and suffer from scalability issues, especially when faced with complex high-dimensional robotic systems or multiagent environments.

Given the connection to safety value functions, it is not entirely surprising that CBF computation methods lose in scalability as they gain in generality. In fact, when continuously differentiable, the safety value function V is itself a valid CBF for safe set Ω^* and can therefore be used in a smooth CBF-type safety filter instead of a switch-type one.

3.3. Rollout-Based Safety Filters: Model Predictive Control

For high-dimensional dynamical systems, computing the optimal value function is prohibitive due to poor scalability, while CBF methods lack general constructive mechanisms. Instead, rollout-based safety filters aim to verify system safety at runtime by forward-simulating (rolling out) the dynamics model or solving a trajectory optimization problem.

3.3.1. Model predictive shielding. The basic model predictive shielding (MPS) algorithm (60) works under the assumption that the dynamical model is fully accurate (no disturbance) and the filter has knowledge of a fallback policy $\pi^\heartsuit$ as well as a terminal safe set Ω (typically small, e.g., at-rest states) that is invariant under $\pi^\heartsuit$. Monitoring is conducted at each control cycle by simply checking if, after executing the candidate task control $\pi^{\text{task}}(x)$, the fallback policy $\pi^\heartsuit$ would safely drive the system state into Ω within the prediction horizon (a sufficient condition for all-time safety, although not generally necessary, and therefore typically conservative). Intervention is then a binary policy switch: If the safety monitor's check is successful, the filter lets the task control $\pi^{\text{task}}(x)$ through; if unsuccessful, it overrides it with the fallback policy $\pi^\heartsuit$, whose ability to preserve safety from x was necessarily verified at an earlier control cycle. This simple scheme illustrates an important feature of safety filters: They preserve (with the corresponding type of guarantee) the robot's future ability to maintain safety using a fallback safety strategy. This results in a recursive safety guarantee: As long as the fallback can maintain safety from the initial state, it will be able to maintain safety from every subsequent one.

While the methodology is agnostic to the choice of fallback policy $\pi^\heartsuit$ and terminal safe set Ω , the conservativeness of the resulting filter depends greatly on them. If poorly constructed, the fallback policy will need to be used often, in order to avoid reaching a state from which it cannot safely reach Ω . Similarly, a small terminal set Ω may be harder to reach by the fallback policy, triggering more frequent interventions. On the opposite extreme, if given the maximal safe set Ω^* and an optimal safety policy π^* , MPS recovers the least-restrictive filter of Equation 11 (since the

safety monitor's validation only fails whenever the proposed action immediately drives the state out of Ω^*).

3.3.2. Forward-reachable sets. When the system model is not perfectly accurate but subject to bounded predictive uncertainty, its future safety can be checked using a forward-reachable set (FRS). Given an initial state and a control policy, the FRS contains all possible future states, including those reached under the worst-case uncertainty realization. Safety is then determined in terms of whether the FRS under the fallback policy eventually becomes fully contained in the terminal safe set Ω without previously intersecting the failure set $\mathcal{F}$.

Reachability-based trajectory design (61, 62) considers a parameterized space of nominal trajectories and associated tracking controllers and precomputes the FRS of possible tracking errors around each nominal trajectory. At runtime, unsafe trajectories (those whose FRS cannot come to a full stop without colliding) are ruled out, and task-driven control optimization is restricted to the remaining set of safe parameter values—an optimization-type intervention scheme similar to the CBF constraint in Equation 13.

Recent research efforts in computing FRSs for verification include high-dimensional dynamics (63), hybrid systems (64), and imperfect observations (65). We refer interested readers to a recent survey of set representations and computation for FRSs (66). Motivated by the growing use of neural network controllers, the study of neural network output robustness (67) has been extended to FRS computation for neural network-controlled systems (68–70). Obtaining the FRS for a predefined fallback policy decouples synthesis from verification, inducing a more tractable computational problem. The downside is that the precomputed policy does not account for the FRS, which can result in conservative filtering and diminished task performance.

3.3.3. Tube-based model predictive control. Tube-based model predictive control (tube MPC) can be used to compute a local fallback policy via runtime optimization at each control cycle, rather than relying on a precomputed one. Inspired by set and viability theory (26, 71), tube MPC optimizes a nominal trajectory alongside a robust FRS (tube) encompassing all possible tracking deviations. Existing techniques include an offline-synthesized tube with its linear auxiliary control policy (72, 73), constraint tightening through contraction theory and sum-of-squares optimization (74), and online tube optimization with min–max differential inequalities (75, 76). A related approach is fast and safe tracking (FaSTrack) (77), which precomputes a tracking error bound and a nonlinear tracking policy by solving an offline pursuit–evasion game between a high-fidelity dynamical model and a lower-fidelity one used for runtime planning.

Tube MPC-based safety filters were first proposed by Wabersich & Zeilinger (78) for linear dynamics with additive uncertainty. At each control cycle, the tube MPC filter solves a runtime program to compute a nominal plan $\hat{x}_k, \hat{u}_k$ with disturbance-free dynamics:

$$\min_{\hat{u}_k \in \hat{\mathcal{U}}} \left\| \hat{u}_0 - \pi^{\text{task}}(x) \right\|^2 \quad 14a.$$

$$\text{subject to} \quad \hat{x}_0 = x, \quad \hat{x}_{k+1} = f(\hat{x}_k, \hat{u}_k, 0), k \in \{0, \dots, H-1\} \hat{x}_{k+1} = f(\hat{x}_k, \hat{u}_k, 0), k \in \{0, \dots, H-1\}, \quad 14b.$$

$$\hat{x}_k \notin \hat{\mathcal{F}}, \quad k \in \{0, \dots, H-1\}, \quad 14c.$$

$$\hat{x}_H \in \hat{\Omega}, \quad 14d.$$

where the adjusted failure set $\hat{\mathcal{F}} \supset \mathcal{F}$, terminal safe set $\hat{\Omega} \subset \Omega$, and control set $\hat{\mathcal{U}} \subset \mathcal{U}$ are computed to account for the tracking error and disturbance rejection effort under $d \in \mathcal{D}$, ensuring that the

Forward-reachable set (FRS): the set of all states x_k that might be reached by a system at time k , given an initial state x_0 and a control policy π , under all possible $d \in \mathcal{D}$

Region of attraction:

the set of all initial states x from which trajectories converge asymptotically to a given equilibrium point x^* under a suitable policy π

entire FRS reaches Ω while clearing $\mathcal{F}$. Monitoring is thus embedded in constraint feasibility, and the intervention is given by the computed solution as $\phi(x, \pi^{\text{task}}(x)) = \hat{u}_0$.⁷

Tube MPC-based safety filters have been extended to handle nonlinear dynamics by sampling (79), statistical bounds (80), and probabilistic reachable sets (81). Additionally, a data-driven dynamics model with state- and input-dependent uncertainty has been considered (2) and can be combined with more learning-based MPC methods. In general, the optimization problem of tube MPC can be computationally intensive, especially for high-dimensional systems and complex constraints. The next section reviews a variety of safety filters that aim to improve scalability by leveraging data and machine learning methods.

3.4. Data-Driven Safety Filters: Deep Learning

Deep learning has proven to be an effective framework to approximately solve optimal control problems using neural network-parameterized value functions (critics) and control policies (actors). Recent work has used deep RL with a binary cost $c(x, u) := \mathbb{1}\{f(x, u, 0) \in \mathcal{F}\}$, learning a safety critic that can be seen as a risk monitor (82, 83). This approach tends to exhibit slow convergence due to the sparse learning signal. In addition, the fact that the algorithm can only learn by experiencing safety failures $x \in \mathcal{F}$ makes online learning impractical for robotic applications where high-fidelity simulators are unavailable (84). In some cases, however, the safety critic can instead be trained offline from prior data, in a postmortem fashion (85).

Another line of research has pursued approximate solutions to the optimal safety problem encoded in Equation 9 (5, 6, 38–40). Unlike sparse binary cost functions, the continuous margin function g provides a dense training signal that additionally allows learning from near-failure events. Self-supervised learning has been employed to approximately solve the safety Bellman equation (40) and Isaacs equation (6, 86) and give probabilistic guarantees (87). These methods often assume that the algorithm has access to an explicit (uncertain) dynamics model and that the min-max optimization in Equation 9 can be solved efficiently at any given state x . In contrast, deep RL approaches require only that the system be available (physically or as a black-box simulator) to collect state-control sequences. Deep RL has similarly been used to approximate the solution to the safety Bellman equation (38, 39, 88) and the zero-sum Isaacs equation (5), as well as to obtain probabilistic guarantees (84). These deep approximation methods show promise in scaling to high-order nonlinear robot dynamics, such as legged locomotion (**Figure 4a**). Finally, runtime Bayesian learning has been used to reassess the robot's uncertainty in light of ongoing observations, updating the computed safety value function and adapting the conservativeness of the associated safety filter (89) (**Figure 4b**).

Closely related to approximating the optimal safety value function, recent efforts aim to learn CBFs from data. Robey et al. (4) introduced an optimization-based method that synthesizes a CBF for nonlinear systems from expert demonstrations and verifies its validity through smoothness-based sufficient conditions. Extensions handle hybrid systems (58) and perception uncertainty (59). A conceptually similar approach (93) learns a neural network-parameterized CBF based on a set of safe and unsafe state samples.

Data-driven stability analysis can be used to learn controlled-invariant regions of attraction for safety assurance. For example, Berkenkamp et al. (94) proposed a safe RL algorithm that yields high-probability stability guarantees during the learning process, using a Lyapunov-based fallback policy π whose learned region of attraction is gradually enlarged as the system identifies its

⁷Similar to MPS, whenever the constrained optimization cannot find a feasible solution, a fallback policy can be obtained from the last feasible cycle. In fact, these prior feasible controls are typically used as initialization to speed up the runtime optimization.

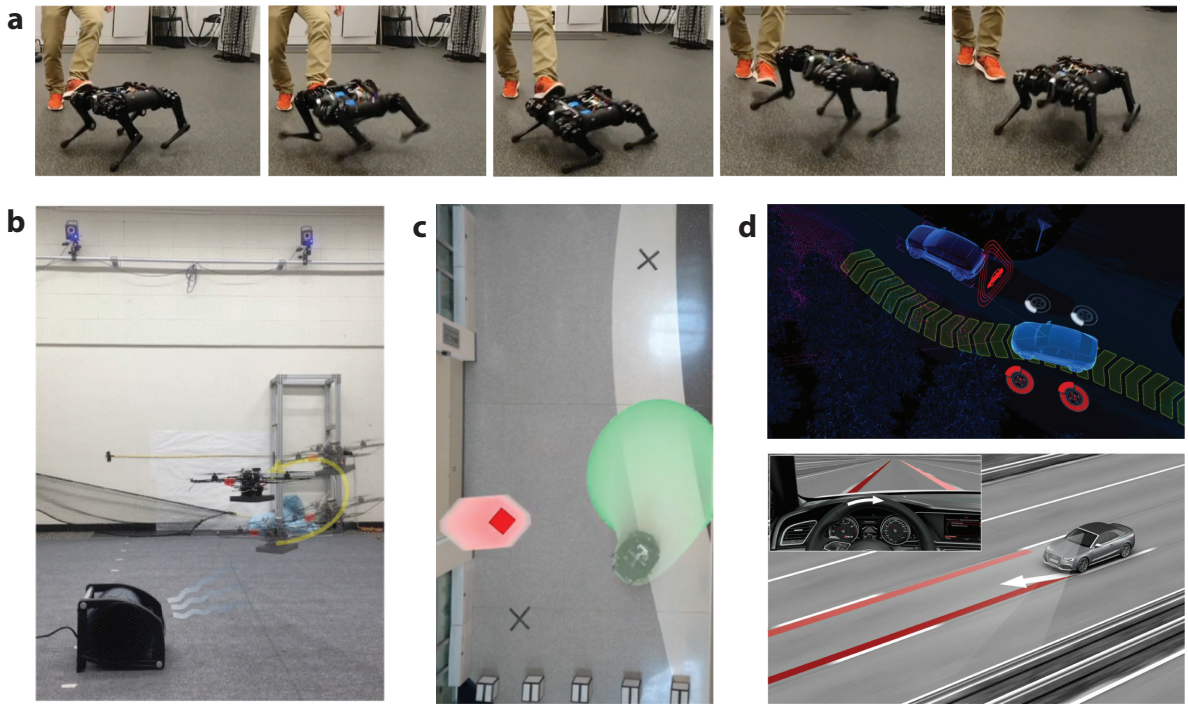


Figure 4

Applications of safety filters. (a) A quadruped robot uses the adversarial RL policy (5) as the safety filter to prevent falling when kicked by a human. (b) A quadrotor learns to hover in the presence of wind disturbance while using an HJI-based fallback policy to ensure safety (89, 90). (c) An indoor mobile robot navigates around a dynamic obstacle (red) using an FRS-based predictive safety filter. (d) Advanced driver assistance systems in modern cars include simple safety filters that automatically engage fallback maneuvers like braking or steering away from detected obstacles to avoid or mitigate imminent collisions. Abbreviations: FRS, forward-reachable set; HJI, Hamilton–Jacobi–Isaacs; RL, reinforcement learning. Panel *b* adapted with permission from Reference 90; copyright 2019 IEEE. Panel *c* adapted with permission from Reference 62. Panel *d* adapted from References 91 (top) and 92 (bottom).

dynamics. The recent stabilize–avoid formulation (95) aims to learn policies that safely drive the state to some equilibrium in a region of interest.

Although neural network critics have shown practical utility in value-based monitoring, it must be kept in mind that the corresponding safety filters come with no guarantees. Using statistical generalization theory, probabilistic guarantees can be provided under the assumption that training environments and the deployment conditions are drawn from the same (unknown) distribution (84). In Section 4.2, we discuss an alternative to learned critics, which combines deep learning synthesis with model predictive monitoring, effectively treating the learned controller as an untrusted oracle to guide the fallback strategy, which can then be verified at runtime through FRS rollouts to obtain robust safety guarantees.

3.5. Human-Centered Safety Filters

Significant efforts have been directed toward the advancement of robotic systems capable of coexisting and engaging with other agents, particularly humans. Enforcing safety becomes especially challenging in interactive settings, as the robot and its peer agents may have coupled dynamics, limited communication capabilities, and conflicting interests. Robust safety methods for multiagent systems consider worst-case agent behavior (96), which can lead to the overly

conservative decision-making known as the freezing robot problem (19), negatively impacting the robot's ability to complete assigned tasks. Several alternatives have been proposed with the aim of unfreezing the robot, to improve task performance without compromising on safety.

3.5.1. Adaptive safety filters. To prevent overly conservative robot behavior based on an a priori worst-case assumption, adaptive safety filters dynamically constrain the disturbance set $\mathcal{D}$, modulating the range of admissible peer agent actions, based on environmental conditions or observed agent behavior. Forward reachability methods (62, 97, 98) use behavior prediction models in combination with reachability analysis to produce robust FRS predictions, which are then used for planning (**Figure 4c**). Althoff & Dolan (65) imposed modeling assumptions on driving behavior informed by traffic rules to yield less conservative FRSs of other road users. Recent work (99–101) has dynamically adjusted the safety filter based on observed human behavior. While these methods provide a tuning knob to balance safety and performance, all-time safety guarantees are usually lost absent additional assumptions on human behavior. On the other hand, these approaches implicitly assume a static understanding of the human's internal state during the entire safety analysis horizon, neglecting the robot's ability to gain information about the human in the future. This can be formally addressed in a safe partially observable stochastic game (102) [or dual control (12, 103)] framework, which is an emerging topic showing some early research progress (18, 104–107).

3.5.2. Probabilistic safety filters. Another commonly adopted safety method that relaxes the worst-case assumption relies on computing a probabilistically safe control policy. Fisac et al. (99) proposed to have the robot maintain a measure of its degree of confidence in a learned human behavior prediction model at runtime. This allows the robot to plan probabilistically safe trajectories accounting for the (possibly varying) accuracy of human motion predictions. Schildbach & Borrelli (108) leveraged scenario MPC (109) to compute safe lane change strategies for autonomous vehicles while accounting for the uncertain traffic environment (e.g., trajectories of other vehicles) encoded by a set of pregenerated scenarios. Chance constraints (110) are incorporated in the MPC problem, which establishes an upper bound on the probability of the ego vehicle violating the safety specifications. Ultimately, however, under any such probabilistic approaches, safety can be compromised when other agents take actions that are assigned low probability by the robot's prediction algorithm [the long tail of unlikely events (23)].

3.5.3. Filter-aware task policies. Seeking a tighter integration between safety and performance, some efforts have investigated endowing task policies with safety filter awareness. These approaches relax the safety–performance separation paradigm, allowing the robot's task-driven decisions to explicitly account for the safety filter's behavior; the goal is to avoid unnecessarily triggering interventions that may negatively impact performance. This idea was explored by Akametalu et al. (89) in the safe learning context, integrating the HJ safety value function V into the RL reward signal, leading to a reduced frequency of safety filter interventions, which was shown to benefit the learning process. More recently, Leung et al. (96) studied filter awareness for safety-critical human–robot interaction in autonomous driving, with an HJ safety filter; Hu et al. (111) developed a similar shielding-aware robust planning (SHARP) framework for a broader class of safety filters, and follow-up work (18) combined filter awareness with active uncertainty reduction to further mitigate the impact of safety overrides under low-probability behaviors of the other agent. Avoiding abrupt safety interventions is also critical in human-centered safety filters such as advanced driver assistance systems (ADAS) in modern cars (**Figure 4d**), where unexpected automatic overrides can create new safety risks.

3.6. Safety Filters for Imperfect Perception

The methods discussed thus far rely on the common assumption that the robot has reliable access to an accurate estimate of all relevant state variables. This assumption is increasingly challenged by modern robotic applications. Various forms of imperfect perception, ranging from occlusions to sensor failures, can significantly endanger the robot's operation. Under probabilistic uncertainty, computing safe control strategies that account for the robot's future sensing ability can be formulated as a partially observable Markov decision process or stochastic game, which is generally intractable and requires approximation (see, e.g., 17, 112–114); the worst-case counterpart of this problem can be tackled using robust reachability analysis. Koschi & Althoff (115) proposed an approach to compute the FRS of occluded objects, considering road structures and traffic rules in driving scenarios. Laine et al. (116) provided an HJ safety analysis framework under the possibility that observations of currently tracked objects may be lost at a future time. These methods conservatively model the robot's future access to information, relying on eyes-closed safety fallback strategies that do not respond to potential future observations.

In contrast, Zhang & Fisac (106) incorporated the robot's measurement model into the safety synthesis to obtain safety strategies that actively seek and adapt to informative observations of currently occluded regions and potential objects. The problem is formulated as a two-mode pursuit–evasion game with an initially occluded pursuer (open-loop phase) that the robot may at some point discover within its field of view and subsequently avoid (closed-loop phase). The framework provides a safety monitoring criterion that verifies whether the robot's strategy will enable it to detect any potential agent currently hidden beyond its field of view before it becomes impossible to avoid a collision with it. This leads to a simple switching safety filter with worst-case guarantees under imperfect observations. Follow-up work (117) planned occlusion-aware trajectories with a learned generative model that reasons about unobserved agents, who may be revealed in the near future, depending on the robot's actions. This learning-based approach enjoys better scalability at the expense of safety guarantees, which unfortunately are no longer available. CBFs (Section 3.2) have also been extended for robust motion planning against imperfect measurements (52, 59).

Safety analysis under imperfect state perception and under uncertain agent intent (Section 3.5.3) share a common technical bottleneck, namely enabling the robot to strategically account for its future ability to acquire new information. Future research should focus on developing a single safety analysis framework that unifies those two emerging fields.

4. UNIFIED SAFETY FILTER THEORY

This section introduces a mathematical formalism that brings a broad class of the safety filters reviewed in this article under a common lens. The exposed modular structure can be used to reconceptualize a range of well-established safety filters, as well as to guide the construction of new ones. The analysis focuses on robust safety filters for uncertainty $d \in \mathcal{D}$, while noting that the concepts and results have analogous probabilistic counterparts.

Definition 1 (safety monitor). Given an (uncertain) system as in Equations 1 and 2 and an available fallback policy $\pi^\nabla : \mathcal{H} \rightarrow \mathcal{U}$, a function $\Delta^\nabla : \mathcal{H} \times \mathcal{U} \rightarrow \mathbb{R}$ is a safety monitor if it satisfies the (one-way) implication

$$\Delta^\nabla(\eta, u) \geq 0 \Rightarrow \eta \notin \mathcal{F}^\eta \wedge f^\eta(\eta, u, d) \in \Omega^\nabla, \forall d \in \mathcal{D}, \quad 15.$$

where $\mathcal{F}^\eta \subseteq \mathcal{H}$ is the (possibly implicit) information-space failure set, containing all information states that do not preclude a current failure $x \in \mathcal{F}$, and $\Omega^\nabla \subseteq \mathcal{H}$ is the (possibly

Information-space failure set: the set $\mathcal{F}^\eta$ of all information states $\eta \in \mathcal{H}$ that allow the possibility of some failure state, i.e., $\mathcal{X}(\eta) \cap \mathcal{F} \neq \emptyset$

Information-space

safe set: the set

$\Omega \subseteq \mathcal{H}$ of information states from which

some policy

$\pi : \mathcal{H} \rightarrow \mathcal{U}$ maintains

all-time safety for all

realizations of the

uncertainty; Ω is

controlled-invariant

(under π) and disjoint

from the

information-space

failure set, i.e.,

$\Omega \cap \mathcal{F}^\eta = \emptyset$

implicit) information-space safe set under π^Ψ , containing all information states η such that π^Ψ enforces Equation 5 from all possible starting states x consistent with η .

Definition 2 (safety filter). A fallback policy $\pi^\Psi : \mathcal{H} \rightarrow \mathcal{U}$, safety monitor $\Delta^\Psi : \mathcal{H} \times \mathcal{U} \rightarrow \mathbb{R}$, and intervention scheme $\phi : \mathcal{H} \times \mathcal{U} \rightarrow \mathcal{U}$ constitute a (robust) safety filter if they jointly satisfy the implication

$$\Delta^\Psi(\eta, \pi^\Psi(\eta)) \geq 0 \Rightarrow \Delta^\Psi(\eta, \phi(\eta, u)) \geq 0, \forall u \in \mathcal{U}. \quad 16.$$

Definition 1 establishes that a safety monitor returns a positive value for a candidate control action only if the known fallback policy is guaranteed to maintain safety after the action is taken. Definition 2 then requires that a (robust) safety filter intervene to preclude any control actions that fail the monitoring check whenever the fallback policy itself passes it. The following general safety filter result then follows directly from the two definitions.

Theorem 1 (universal safety filter theorem). Consider an (uncertain) system as in Equations 1 and 2 and a safety filter as in Definition 2, comprising a fallback policy π^Ψ , a safety monitor Δ^Ψ as in Definition 1, and an intervention scheme ϕ . If the system is deployed at an initial information state $\eta_0 \in \mathcal{H}$ deemed safe by the safety monitor under the fallback policy, i.e., $\Delta^\Psi(\eta_0, \pi^\Psi(\eta_0)) \geq 0$, then its evolution under filtered dynamics $f^\phi(x, u, d) := f(x, \phi(\eta, u), d)$ satisfies the safety condition in Equation 5 for all possible task policies $\pi^{\text{task}} : \mathcal{H} \rightarrow \mathcal{U}$.

The theorem's result relies on the fact that the safety filter preserves the positivity of the safety monitor's checks, and this in turn guarantees that it is always possible to robustly maintain all-time safety by resorting to the fallback policy. The safety filter may never actually apply the fallback policy and instead merely conserve the option to use it at a later time.

We now showcase, through a nonexhaustive set of example corollaries, how safety guarantees for various classes of safety filters can be readily rederived as special instances of Theorem 1. For fully observable settings, we simply let $\eta \equiv x$.

Corollary 1 (least-restrictive safety filter). The HJ safety filter with $V : \mathcal{X} \rightarrow \mathbb{R}$, the maximal safe set $\Omega^* \subset \mathcal{X} \subseteq \mathbb{R}^{n_x}$, and the intervention scheme ϕ defined in Equation 11 maintains all-time safety from any initially safe state $x_0 \in \Omega^*$.

Proof. Let the safety monitor be $\Delta^\Psi(x, u) := \min_{d \in \mathcal{D}} V(f(x, u, d))$ and the fallback policy be $\pi^\Psi(x) := \pi^*(x)$. The filter based on Equation 11 then satisfies Definitions 1 and 2, and initially $\Delta^\Psi(x_0, \pi^*(x_0)) \geq 0, \forall x_0 \in \Omega^*$. Applying Theorem 1 completes the proof. $\square$

Corollary 2 (CBFs). The CBF safety filter with CBF $b : \mathcal{X} \rightarrow \mathbb{R}$, with $\alpha(a) < a/\Delta t, \forall a \in \mathbb{R}$, safe set $\Omega \subset \mathcal{X} \subseteq \mathbb{R}^{n_x}$, and intervention scheme ϕ defined as the CBF-QP in Equation 13, maintains all-time safety from any initially safe state $x_0 \in \Omega$.

Proof. Let the safety monitor be $\Delta^\Psi(x, u) := \min\{b(x), \dot{b}(x, u) + \alpha(b(x))\}$ and the fallback policy (implicitly defined) be $\pi^\Psi(x) := \arg \max_u \dot{b}(x, u)$. Indeed, if $\Delta^\Psi(x, u) > 0$, then it must be that $\dot{b}(x, u) \geq -\alpha(b(x))$. Then, at the next control cycle,⁸ $b(x') \geq b(x) - \alpha(b(x))\Delta t > b(x) - b(x) = 0$, so $x' \in \Omega$, satisfying Definition 1. On the other hand, if $\Delta^\Psi(x, \pi^\Psi(x)) \geq 0$, then this implies that the CBF-QP is feasible, and therefore $\Delta^\Psi(x, \phi(x, u)) \geq 0$ for any $u \in \mathcal{U}$,

⁸Since the CBF formulation assumes continuous-time control, we neglect the integration error in b between control cycles. Practical CBF filter implementations typically enforce $\dot{b} \geq \epsilon > 0$.

satisfying Definition 2. Noting that $\Delta^\Psi(x_0, \pi^\Psi(x_0)) \geq 0, \forall x_0 \in \Omega$, and applying Theorem 1 completes the proof. $\square$

The next corollary combines MPS (Section 3.3.1) with FRSs (Section 3.3.2).

Corollary 3 (robust MPS). Consider a rollout safety filter with available fallback policy π^Ψ , known terminal safe set Ω , and switch-type intervention scheme

$$\phi(x, u) = \begin{cases} u, & \hat{\mathcal{X}}_\tau \cap \mathcal{F} = \emptyset, \forall \tau \in \{0, \dots, H\} \wedge \hat{\mathcal{X}}_H \subseteq \Omega, \\ \pi^\Psi(x), & \text{otherwise,} \end{cases} \quad 17.$$

with $\hat{\mathcal{X}}_0 = \{x\}$, $\hat{\mathcal{X}}_1 = \{f(x, u, d), d \in \mathcal{D}\}$, and $\hat{\mathcal{X}}_{\tau+1} = \{f(\hat{x}, \pi^\Psi(\hat{x}), d), \hat{x} \in \hat{\mathcal{X}}_\tau, d \in \mathcal{D}\}, \tau \geq 1$. This filter maintains all-time safety from any deployment state $x_0 \in \Omega$.

Proof. Let $\Delta^\Psi(x, u) := \mathbb{1}\{\hat{\mathcal{X}}_\tau \cap \mathcal{F} = \emptyset, \forall \tau \in \{0, \dots, H\} \wedge \hat{\mathcal{X}}_H \subseteq \Omega\} - \frac{1}{2}$ be the safety monitor. Since the condition inside of the indicator $\mathbb{1}\{\cdot\}$ is sufficient for all-time safety (π^Ψ drives all possible system trajectories safely into Ω , which is controlled-invariant), Definition 1 is satisfied. Since the intervention scheme applies u only when $\Delta^\Psi(x, u) \geq 0$ and otherwise overrides it with $\pi^\Psi(x)$, the condition $\Delta^\Psi(x, \pi^\Psi(x)) \geq 0$ ensures that $\Delta^\Psi(x, \phi(x, u)) \geq 0$, meeting Definition 2. Noting $\Delta^\Psi(x, \pi^\Psi(x)) \geq 0, \forall x \in \Omega$, and applying Theorem 1 completes the proof. $\square$

We now consider a safety filter for the imperfect-information setting of environment navigation with sensor occlusion, as discussed in Section 3.6.

Corollary 4 (model predictive safety filter for unmapped environments). Consider a mobile robot with perfect estimation of its own state x^R and known discrete-time dynamics f^R navigating a static but initially unmapped environment $\mathcal{W} \subseteq \mathbb{R}^3$. The robot is equipped with an onboard perception stack that conservatively identifies adjacent free regions and adds them to the known free space $\mathcal{W}^{\text{free}}(\eta_k) \subseteq \mathcal{W}$ at each time k . The model predictive safety filter that (a) rejects any actions that would preclude the robot from coming to a full stop in H steps within the currently known free space using a simple braking fallback policy $\pi^\Psi(\eta) \equiv u^{\text{brake}}$ and (b) overrides them with u^{brake} maintains safety from any deployment information state η_0 that has the robot at rest within the initially known free space.

Proof. In this case, the failure set $\mathcal{F}^\eta$ contains any information states η such that the robot's (known) position is outside of $\mathcal{W}^{\text{free}}(\eta)$ and therefore may be in collision. Let $\Delta^\Psi(\eta, u)$ return +1 if π^Ψ brings the robot to a full stop from the next physical state $f^R(x^R, u)$ within another $H - 1$ steps while remaining in the currently known free space $\mathcal{W}^{\text{free}}(\eta)$, and -1 otherwise. We can see that $\Delta^\Psi(\eta, u) = +1$ implies that the robot is currently inside $\mathcal{W}^{\text{free}}(\eta)$, or equivalently $\eta \notin \mathcal{F}^\eta$; since, additionally, $\mathcal{W}^{\text{free}}(\eta_{k+1}) \supseteq \mathcal{W}^{\text{free}}(\eta_k)$ (the robot can still stop safely after any new free space is discovered), Definition 1 is satisfied. The simple filter override with u^{brake} satisfies Definition 2 by construction, and Theorem 1 completes the proof. $\square$

4.1. Safety Value Functions as Control Barrier Functions

CBFs provide a smooth intervention mechanism at the cost of settling for suboptimal safe sets and lacking general synthesis methods. Exploiting the modular structure of safety filters, a promising research direction is to combine CBF-type intervention with other safety approaches that enjoy principled constructive mechanisms. Ongoing research efforts explore the use of the HJ safety value function as a CBF. Choi et al. (118) bridged the dynamic programming safety formulation

and the CBF condition to define and compute control barrier value functions. Similar principles allow CBF candidates to be fine-tuned online (119). Most recently, Kumar et al. (120) proposed to locally approximate the safety value function by runtime differential dynamic programming, constructing an implicit CBF (121, 122) whose (also implicit) safe set approximates Ω^* .

4.2. Offline Neural Synthesis with Runtime Certification

Model predictive safety filters (Section 3.3; Corollary 4) are designed to enforce safety using an arbitrary fallback policy: The more effective the fallback is, the more freedom the filter can afford to grant. This creates a natural synergy with modern neural safety policies trained by RL (Section 3.4), which can be effective in practice but typically lack guarantees. Bastani et al.'s (80) statistical MPS uses RL to train a fallback policy to reach a known terminal safe set. By sampling a number of disturbance sequences for simulation, this method computes a statistical union bound for probabilistic guarantees. Hsu et al. (5) proposed cotraining a safety controller and a worst-case disturbance policy through adversarial RL, approximately solving the Isaacs equation shown in Equation 9. Certification is then provided at runtime by robust rollout-based monitoring, at the cost of some added conservativeness incurred by the FRS computation.

5. DISCUSSION

Advancements in safe decision-making have been accelerating in recent years to meet the increasingly challenging demands of robot autonomy. In this review, we have provided a unified perspective on a growing family of task-agnostic tools that filter arbitrary control policies into safe ones, exposing their shared structure and comparing their main features. We conclude by briefly discussing new research opportunities in safety-enforcing control in the Future Issues list below.

FUTURE ISSUES

1. Safety filters that account for runtime learning: The design of safety filters for settings involving partial observability or agents with unknown intent usually assumes a static information state, leading to overly conservative monitoring and intervention. Pushing the limit of safe autonomy in the information space calls for safety filters that tractably account for the system's runtime learning capabilities.
2. Safety filters that account for interaction: Many safety filters assume nonresponsive or actively adversarial behavior of other agents in the environment, to avoid precariously relying on the actions of others to ensure safety. Delineating practically acceptable assumptions for less conservative, yet no less meaningful, safety assurances in interactive contexts like autonomous driving and robotic caregiving is likely to become a technical and regulatory priority in the coming years.
3. Safety filters with high-dimensional representations: Modern robot autonomy frameworks increasingly rely on high-dimensional sensory input and learned latent representations. Reconciling these frameworks with the need for rigorous safety guarantees grounded in dynamical models is an open and important research challenge.

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